ESE 123 Lab 6 Report: Operational Amplifiers

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Objective. Build and test an inverting amplifier using an LM741 op amp, measure its output limits, and test the project's transimpedance (current-sense) amplifier.

1. Pre-lab design

Inverting amplifier, target gain −4.7: $A_v = -R_f/R_{in}$
Chosen standard values: R_{in} = R1 = 10 kΩ, R_f = R2 = 47 kΩ ⟶ A_v = −47k/10k = −4.7

Project non-inverting amplifier (dc gain): $A_v = 1 + R12/R13 = 1 + 150k/4.75k = 32.58$
Low-pass corner from R12 and C7: $f_c = 1/(2\pi \cdot 150\ k\Omega \cdot 100\ nF) \approx 10.6$ Hz (calculated)

2. Inverting amplifier measurements (±15 V supplies, 1 kHz triangle wave)

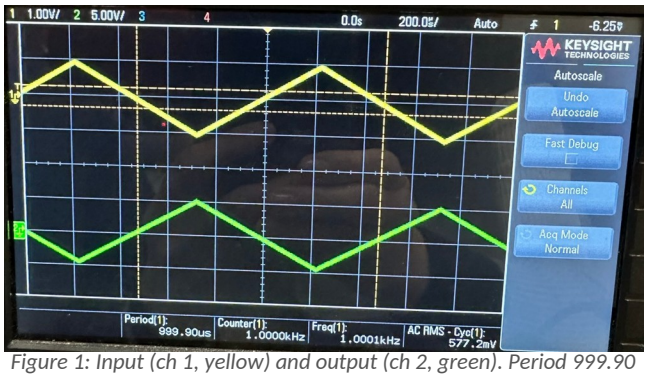


Figure 1: Input (ch 1, yellow) and output (ch 2, green). Period 999.90 µs, f = 1.0001 kHz, input AC RMS 577.2 mV.

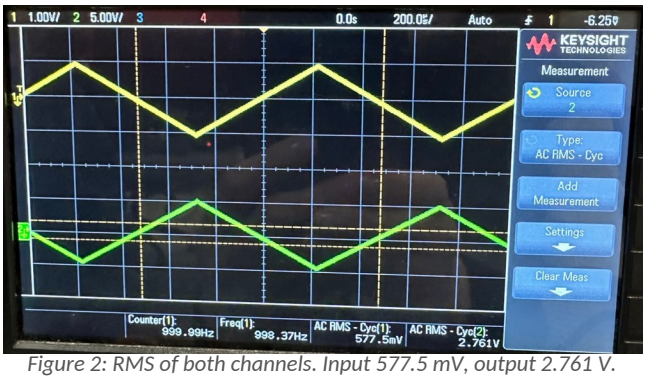


Figure 2: RMS of both channels. Input 577.5 mV, output 2.761 V.

Quantity	Value
Frequency (counter)	999.99 Hz
Input RMS (channel 1)	577.5 mV
Output RMS (channel 2)	2.761 V (inverted)
Measured gain	−2.761 / 0.5775 = **−4.78**
Design gain / % error	−4.70 / +1.7 %

The output is an inverted copy of the input: when the input rises, the output falls. This is expected for an inverting amplifier. The input reads 577.5 mV RMS, about 2 V_{pp}, instead of the 1 V_{pp} setting. That's because the generator assumes a 50 Ω load, but the op amp circuit is a high-impedance load. For a triangle wave, $V_{rms} = V_{pp}/(2\sqrt{3}) = 2/(2\sqrt{3}) = 0.577$ V, which matches. The 1.7% gain error is within the tolerance of 5% resistors.

3. Maximum and minimum output voltage

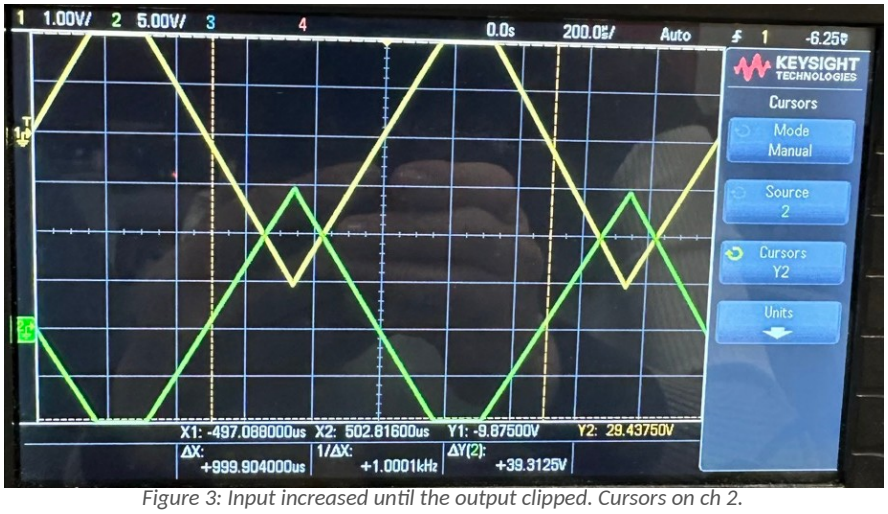


Figure 3: Input increased until the output clipped. Cursors on ch 2.

Cursor (ch 2)	Reading
Y1	−9.875 V
Y2	+29.44 V
ΔY	39.31 V

As I increased the input, the green output trace developed a flat bottom. However, these cursor readings are not a valid measurement of the op amp's output limits. The 39.3 V span is larger than the whole 30 V supply (+15 V to −15 V), so the op amp cannot produce it. Y2 is not sitting on the top of the waveform, and the flat bottom lies on the edge of the screen, so the display may be cutting it off. To repeat this, I should move the channel 2 trace to the center (or use 10 V/div) so both flat spots are on screen, then place Y1 and Y2 on them. For a 741 on ±15 V, I expect roughly ±13 V.

4. Transimpedance amplifier (R11 = 0.07 Ω installed on the board)

Checks before applying current: VDD to ground = 5 V; XTP with no current = 0.8172 mV (close to zero, as expected).

Supply current	V_{XTP} measured	V_{XTP} / I	Expected (2.281 Ω · I)
250 mA	34.06 mV	0.136 Ω	570 mV
500 mA	75.41 mV	0.151 Ω	1.14 V
750 mA	111.5 mV	0.149 Ω	1.71 V
1000 mA	126 mV	0.126 Ω	2.28 V

Discussion. V_{XTP} increases with current, so the sense circuit responds. However, the measured transimpedance (about 0.14 Ω) is far below the designed 2.281 Ω (= 0.07 Ω × 32.58). The amplifier stage doesn't seem to be giving its gain of 32.58. Next, I'll check the U2 solder joints, R12/R13, and the supply's actual current at each step.